1- What is the difference between Cache and Static memory?

The technology of Cache is supposed to be the same as the technology of the CPU. Also, caches are content addressable memory. Then static memory is stored on the CPU

2-Cache memory is a small, high-speed storage area located close to the CPU (external cache) or inside the CPU (Internal Cache).

1) Right 2) Wrong

3-The access time of internal cache is more than external cache.

1) Right 2) Wrong

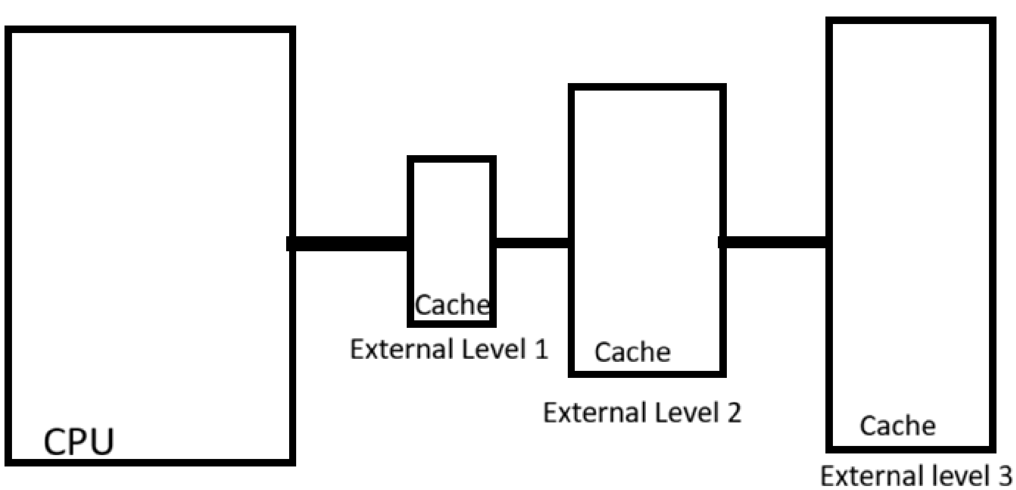
4-It is organized in levels (L1, L2, L3), with L1 being the fastest and smallest.

1) Right 2) Wrong

5-Effective cache management can significantly increase latency and increase processing efficiency.

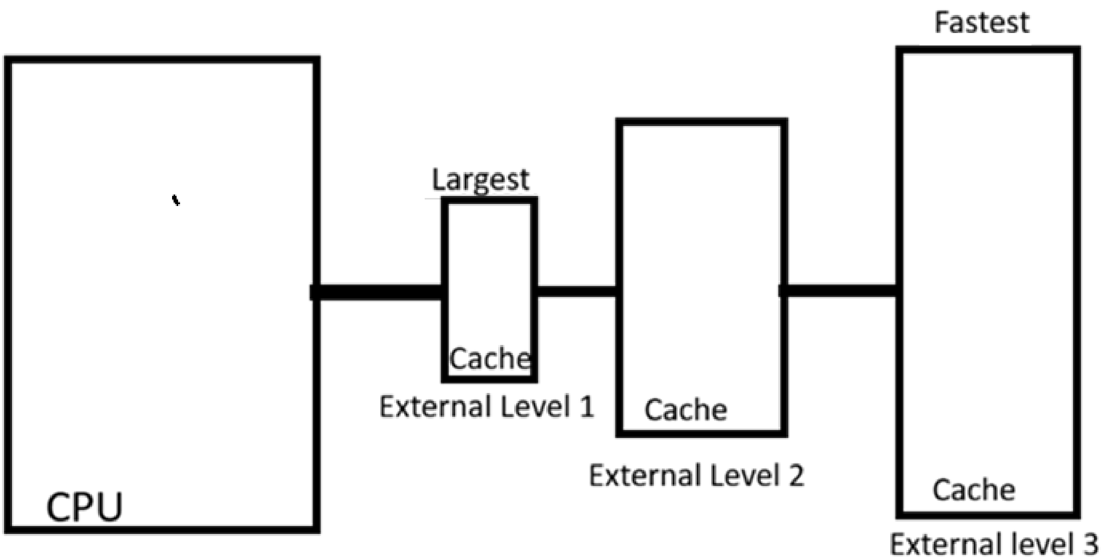
1) Right 2) Wrong

6-This picture represents 3 levels of external cache.



1) Right   2) Wrong

7-What is wrong?



External level 1 is supposed to be the fastest, and external level 3 is the largest.

8-***<u>Core5:</u>*** The cache size varies by generation and model of the Core i5.

- Typically, Core i5 processors have a Level 1 (L1) cache of 32KB per core.
- The Level 2 (L2) cache is usually 256KB per core.
- The Level 3 (L3) cache generally ranges from 3MB to 12MB shared among all cores.
- Newer generations may feature larger caches for improved performance.

1) Right   2) Wrong

9-The Cortex-X3 architecture supports various cache sizes depending on the implementation.

- L1 cache typically ranges from 16KB to 128KB per core.
- L2 cache sizes can vary from 256KB to 1MB per core.
- L3 cache, if present, can range from 2MB to 16MB shared among cores.
- Cache sizes are configurable based on specific use cases and performance requirements.

1) Right 2) Wrong

10-Access time refers to the duration it takes to retrieve data from cache memory.

1) Right 2) Wrong

11-Access time only includes the time to transfer the data.

1) Right 2) Wrong

12-Shorter access times allow the CPU to process instructions more efficiently.

1) Right 2) Wrong

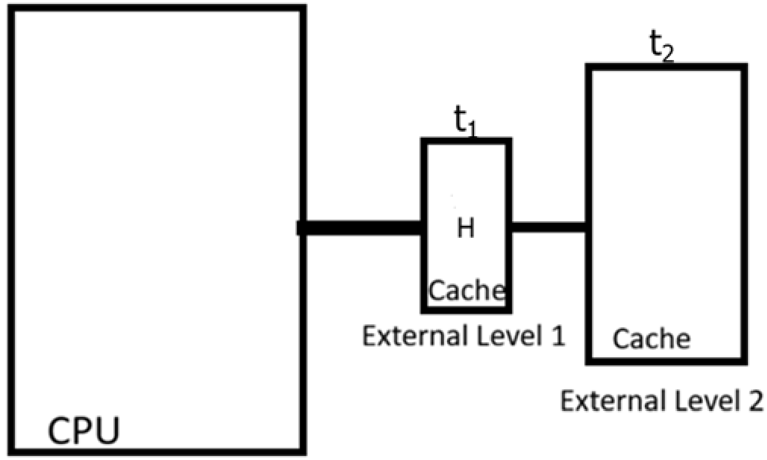
13-Cache Hit vs. Miss: Access time is significantly higher for cache hits compared to cache misses.

1) Right 2) Wrong

14-Access time = Hit time + (Miss rate × Miss penalty).
1) Right 2) Wrong

15-Increasing miss rate and miss penalty can significantly improve overall access time.
1) Right 2) Wrong

16-Explain:

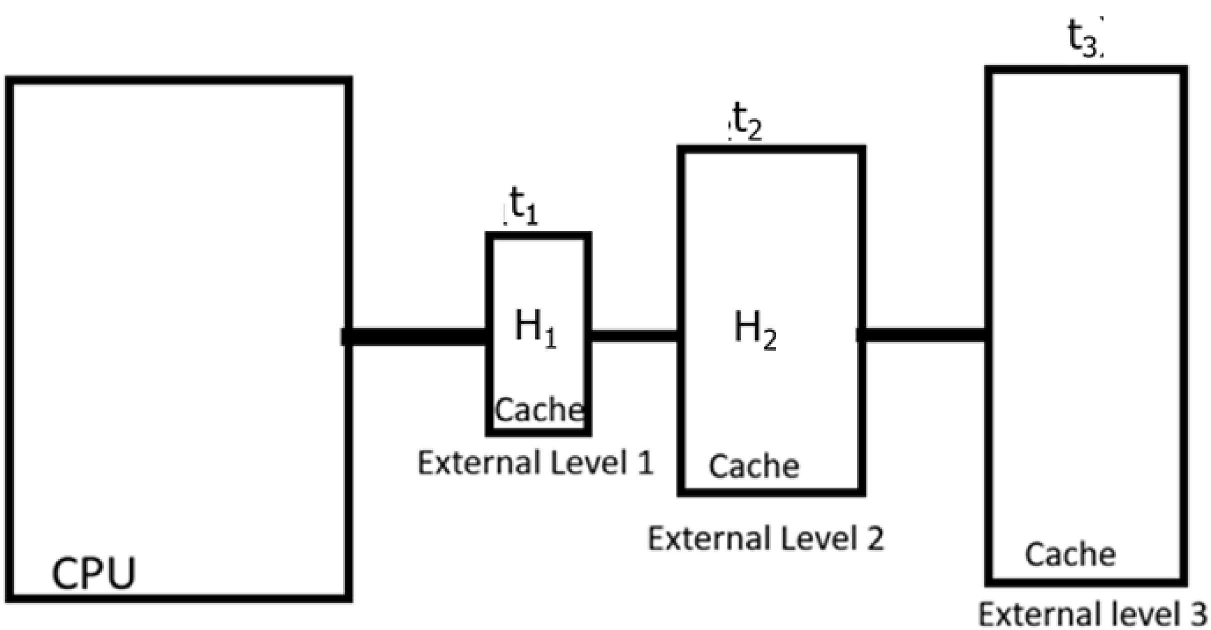


t_1 :Level 1 Cache Access Time
t_2 :Level 2 Cache Access time
H: Hit Ratio

To find the average access time of 2 levels of caches use the formula:

$$t_{avg} = Ht_1 + (1 - H_1)\, t_2$$

17-What is the average access time?



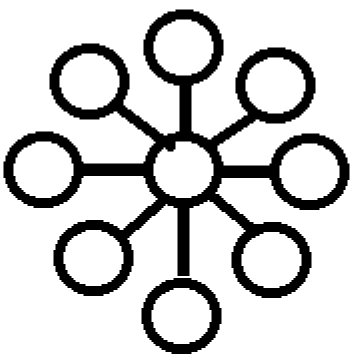
$$t_{avg} = H_1t_1 + (1 - H_1)(H_2t_2 + (1 - H_2)t_3)$$

18-**Multicore Processor:** A single CPU chip with multiple independent cores (execution units) on it, allowing it to run different parts of a single program or multiple programs simultaneously on one chip.
1) Right 2) Wrong

19-**Cluster:** A system with two or more separate physical CPUs on the same motherboard, each with its own cores, sharing memory and resources.

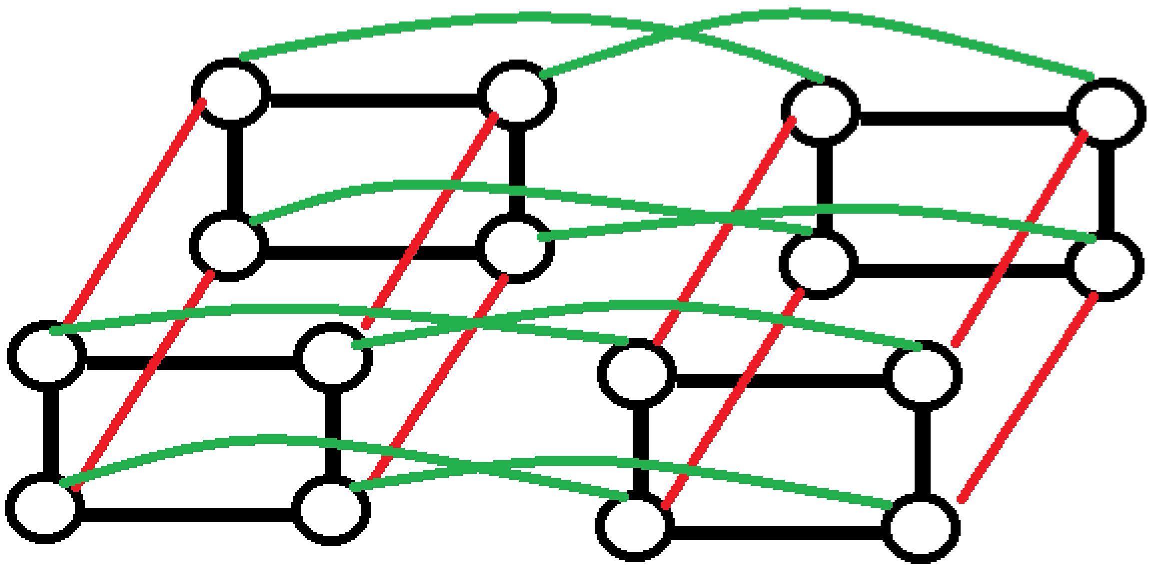
1) Right 2) Wrong

20-This picture illustrates Complete interconnection network:



1) Right 2) Wrong

21-This picture illustrates Mesh interconnection network.



1) Right 2) Wrong